

→ Previous videos

- * Introduced Isabelle interface
- * how to write terms/theorems in Isabelle
- * Two basic proof/reasoning approaches/
 - FW
 - BW

→ This video: Proofs by substitution

- * Suppose we have a then saying

$$\text{lem 1 } \vdash ?x + ?y = ?y + ?x \} \rightarrow \text{simplification rule}$$

&

we have a goal

$$\text{goal: } 1 + 3 \geq 3 + 1$$

Then we can substitute that lemma, rather goal, & we get a new goal

$$\text{goal: } 3 + 1 \geq 3 + 1$$

- * When does it make sense?

• If the R.H.S. of the simpl rule is simpler than the L.H.S., or even if we have a proof R.H.S.

→ Example :-

$$\vdash (x+y) \cdot (a+b) = x \cdot a + x \cdot b + y \cdot a + y \cdot b$$

Note: all are nat's.

$$1. (x+y) \cdot (a+b) = (x+y) \cdot a + (x+y) \cdot b$$

$$\text{using } ?x \cdot (?a + ?b) = ?x \cdot a + ?x \cdot ?b \quad \text{lem 1}$$

$$2. \underline{(x+y) \cdot a} + (x+y) \cdot b = x \cdot a + y \cdot a + (x+y) \cdot b$$

$$\text{using } \text{lem } ? (?a + ?b) \cdot ?x = ?a \cdot ?x + ?b \cdot ?x$$

$$3. \dots = x \cdot a + y \cdot a + x \cdot b + y \cdot b$$

using lem ? ✓

→ Isabelle Notes:-

- * show how to perform the proof

- * find lemmas that are in Isabelle's library

- * a goal of the form $a = ?$ can be proved using ..

- * $a + b + c$ needs as $a + (b + (\underline{c}))$

- * if lem1: $x = y$, then lem1 [symmetric]: $y = x$

- * If the goal is $P \vdash P$, then you close it by ..

- * you can use $x = y$.

also .. = z

also .. = a

also .. = b

finally $x = b$